

수열의 극한

유한 이) 수열의 항은 ∞ 또는 $-\infty$

1) $a_n = -1000/n^2 \rightarrow 0$ 수렴

2) $a_n = 5 - (3/n) \rightarrow 5$ 수렴

3) $b_n = \log_2 n \rightarrow \infty$ 발산

4) $c_n = (-1)^n + 1 \rightarrow 0, 2, 0, 2, \dots$ 발산

5) $a_n = \frac{2n+1}{n+3} \rightarrow 2$ 수렴

유형 02) 극한의 기법

6) $\lim_{n \rightarrow \infty} (1 - \frac{2}{n})(3 - \frac{1}{n^2}) = 3$

7) $\lim_{n \rightarrow \infty} \frac{2n^3+100}{5n^3-1} = \frac{2}{5}$

8) $\lim_{n \rightarrow \infty} \frac{3n^2-n+1}{2n^2+5} = \frac{3}{2}$

9) $\lim_{n \rightarrow \infty} \frac{n+1}{n^2+2} = \frac{1}{\infty} = 0$

10) $\lim_{n \rightarrow \infty} \frac{4n^2+3n}{2n^2-n+1} = \frac{4}{2} = 2$

유형 03) 무한대의 극한

11) $\lim_{n \rightarrow \infty} \sqrt{n^2-2n} - n$
 $= (\sqrt{n^2-2n} - n)(\sqrt{n^2-2n} + n) = n^2-2n-n^2$

$= \frac{-2n}{\sqrt{n^2-2n} + n} = \frac{-2}{1+1} = -1$

12) $\lim_{n \rightarrow \infty} \sqrt{n^2+3n} - n$

$= \frac{(\sqrt{n^2+3n}-n)(\sqrt{n^2+3n}+n)}{\sqrt{n^2+3n}+n} = \frac{n^2+3n-n^2}{\sqrt{n^2+3n}+n} = \frac{3n}{\sqrt{n^2+3n}+n}$

$= \frac{3}{1+1} = \frac{3}{2}$

13) $\lim_{n \rightarrow \infty} (\sqrt{9n^2+n} - 3n)$

$= \frac{(\sqrt{9n^2+n}-3n)(\sqrt{9n^2+n}+3n)}{\sqrt{9n^2+n}+3n} = \frac{9n^2+n-9n^2}{\sqrt{9n^2+n}+3n} = \frac{n}{\sqrt{9n^2+n}+3n}$

$= \frac{1}{3+3} = \frac{1}{6}$

14) $\lim_{n \rightarrow \infty} (\sqrt{n^2+5n} - \sqrt{n^2+n})$

$= \frac{(\sqrt{n^2+5n}-\sqrt{n^2+n})(\sqrt{n^2+5n}+\sqrt{n^2+n})}{\sqrt{n^2+5n}+\sqrt{n^2+n}} = \frac{n^2+5n-n^2-n}{\sqrt{n^2+5n}+\sqrt{n^2+n}} = \frac{4n}{\sqrt{n^2+5n}+\sqrt{n^2+n}}$

$= \frac{4}{1+1} = 2$

15) $\lim_{n \rightarrow \infty} \sqrt{n}(\sqrt{n+1} - \sqrt{n})$

$= \frac{\sqrt{n}(\sqrt{n+1}-\sqrt{n})(\sqrt{n+1}+\sqrt{n})}{\sqrt{n+1}+\sqrt{n}} = \frac{\sqrt{n}(n+1-n)}{\sqrt{n+1}+\sqrt{n}}$

$= \frac{\sqrt{n}}{\sqrt{n+1}+\sqrt{n}} = \frac{n}{2n+1} = \frac{1}{2}$

유형 04) 삼각함수의 극한

16) $\lim_{n \rightarrow \infty} \frac{1}{n} \sin 2n$

$= -1 \leq \sin 2n \leq 1$

$= -\frac{1}{n} \leq \frac{1}{n} \sin 2n \leq \frac{1}{n}$

$\rightarrow 0$

17) $\lim_{n \rightarrow \infty} \frac{\cos n}{n^2}$

$= -1 \leq \cos n \leq 1$

$= -\frac{1}{n^2} \leq \frac{\cos n}{n^2} \leq \frac{1}{n^2}$

$\rightarrow 0$

18) $\lim_{n \rightarrow \infty} \frac{2n+\cos n}{n+1}$

$= -1 \leq \cos n \leq 1$

$= 2n-1 \leq \cos n + 2n \leq 2n+1$

$= \frac{2n-1}{n+1} \leq \frac{\cos n + 2n}{n+1} \leq \frac{2n+1}{n+1}$

$2 \leq \frac{\cos n + 2n}{n+1} \leq 2 \rightarrow 2$

19) $\lim_{n \rightarrow \infty} (1/n^2)(\sin 1 + \sin 2 + \dots + \sin n)$

$\rightarrow 0$

20) $\lim_{n \rightarrow \infty} \frac{n+(-1)^n}{n} = \frac{n+1}{n}, \frac{n-1}{n} = 1$

유형 05) 등비수열의 극한

21) $a_n = \frac{7^n}{4n+5} = \infty$

22) $b_n = \frac{4n^2+5 \cdot 7^{n+1}}{2 \cdot 3^n + 7^n} = \frac{5 \cdot 7^n \cdot 7}{7^n} = 35$

23) $a_n = \frac{5^n - 7^n}{7^n(11^n)} = 0$

24) $b_n = \frac{7^{n+3}}{3^n} = \infty$

25) $a_n = \frac{2^{n+3}}{2^n - 2^n} = 1$

유형 06) 극한값이 주어질 때

26) $\lim_{n \rightarrow \infty} \frac{an+3}{2n-1} = 4 \quad a=8$ (비교항의 계수)

27) $\lim_{n \rightarrow \infty} \frac{3n^2+an}{n^2+1} = 3 \quad a=4$ (우변 상수)

28) $\lim_{n \rightarrow \infty} \sqrt{n^2+an} - n = 3$

$\frac{n^2+an-n^2}{\sqrt{n^2+an}+n} = \frac{an}{\sqrt{n^2+an}+n} = \frac{a}{2} = 6$

* 29) $\lim_{n \rightarrow \infty} \frac{an^2 + n}{2n + 2} = 2$, a 는 존재하지 않는다.

30) $\lim_{n \rightarrow \infty} \frac{2 \cdot 3^n + a}{2n + 1} = 2$, a 의 값에 의존하지 않는다

유형 7) 급수의 수렴, 발산, 역판

31) $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$

① 대입하기

$$= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) \cdots \left(\frac{1}{n} - \frac{1}{n+1}\right)$$

$$= 1 - \frac{1}{n+1}$$

② 극한구하기

$\lim_{n \rightarrow \infty} S_n = 1$, 수렴

32) $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right) =$

① 대입하기

$$= \left(1 - \frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}}\right) + \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}}\right) \cdots \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right)$$

$$= 1 - \frac{1}{\sqrt{n+1}}$$

② 극한구하기

$\lim_{n \rightarrow \infty} S_n = 1$, 수렴

33) $\sum_{n=1}^{\infty} (\sqrt{n} - \sqrt{n+1})$

① 대입하기

$$= (1 - \sqrt{2}) + (\sqrt{2} - \sqrt{3}) + (\sqrt{3} - \sqrt{4}) \cdots (\sqrt{n} - \sqrt{n+1})$$

$$= 1 - \sqrt{n+1}$$

② 극한구하기

$\lim_{n \rightarrow \infty} S_n = -\infty$, 발산

* 34) $\sum_{n=1}^{\infty} \frac{4n-1}{5n+2} = \frac{4}{5}$

* 35) $\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)} = \frac{1}{4n^2-1} = 0$ 이므로 수렴

① 대입하기

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \cdots + \frac{1}{4n^2-1}$$

② 극한구하기

③ 수렴 판정

유형 8 등비 급수의 합 $= \frac{a}{1-r}$
 36) $1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \cdots = \frac{2^{n-1}}{3^{n-1}} = \left(\frac{2}{3}\right)^{n-1}$

① 수렴 판정하기

첫째항: 1, 공비: $\frac{2}{3}$

② 합구하기

$$\frac{1}{1 - \frac{2}{3}} = \frac{1}{\frac{1}{3}} = 3$$

* 37) $\sum_{n=1}^{\infty} (1.01)^n$

첫째항: 1.01, 공비: 1.01

공비 ≥ 1 이므로 발산

38) $\sum_{n=1}^{\infty} \frac{4^{n+1} + (-3)^n}{7^n} = 4 \left(\frac{4}{7}\right)^n + \left(\frac{-3}{7}\right)^n$ (첫째항: $\frac{16}{7}$, 공비: $\frac{4}{7}$)

합: $\frac{\frac{16}{7}}{1 - \frac{4}{7}} = \frac{\frac{16}{7}}{\frac{3}{7}} = \frac{16}{3}$
 $\frac{1}{1 - \frac{3}{7}} = \frac{7}{4}$
 $\frac{21}{4} = \frac{21}{4}$
 $\frac{21}{4} + \frac{2}{7} = \frac{147}{28} + \frac{8}{28} = \frac{155}{28}$

* 39) $\sum_{n=1}^{\infty} \frac{3^n + 2}{7^n} = \left(\frac{3}{7}\right)^n + \frac{2}{7^n}$

① $\frac{3^n}{7^n}$ (첫째항: $\frac{3}{7}$, 공비: $\frac{3}{7}$)

$\rightarrow$ 합 $\frac{\frac{3}{7}}{1 - \frac{3}{7}} = \frac{\frac{3}{7}}{\frac{4}{7}} = \frac{3}{4}$

발산 판정

? $\rightarrow$ ② $\left(\frac{1}{7^n}\right) \cdot 2$ (첫째항: $\frac{2}{7}$, 공비: $\frac{1}{7}$)

$\rightarrow$ 합: $\frac{\frac{2}{7}}{1 - \frac{1}{7}} = \frac{\frac{2}{7}}{\frac{6}{7}} = \frac{14}{42} = \frac{1}{3}$

③ 합 $\frac{3}{4} + \frac{1}{3} = \frac{9+4}{12} = \frac{13}{12}$

40) $\sum_{n=1}^{\infty} \frac{2^n + 3}{6^n} = \left(\frac{2}{6}\right)^n + \frac{3}{6^n}$

① $\left(\frac{2}{6}\right)^n$ (첫째항: $\frac{1}{3}$, 공비: $\frac{1}{3}$)

합: $\frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{3}{6} = \frac{1}{2}$

② $3 \cdot \left(\frac{1}{6}\right)^n$ (첫째항: $\frac{1}{2}$, 공비: $\frac{1}{6}$)

합: $\frac{\frac{1}{2}}{1 - \frac{1}{6}} = \frac{\frac{1}{2}}{\frac{5}{6}} = \frac{6}{10} = \frac{3}{5}$

③ 합 $\frac{1}{2} + \frac{3}{5} = \frac{5+6}{10} = \frac{11}{10}$